Deep Graph Wavelet Networks for Pediatric Seizure Prediction

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Abstract: We present an autonomous study on graph convolutional transformers for multi-channel scalp EE

Introduction: Early detection of epileptic seizures is a paramount clinical challenge in neurology and patien

Methodology: We designed a Spatio-Temporal Graph Neural Network with continuous wavelet spectrogra

Datasets: The model was evaluated on the CHB-MIT Scalp EEG database comprising 24 pediatric patient

Results: The proposed architecture achieved 98.4% sensitivity with a false alarm rate of 0.08 per hour und

Discussion: Graph convolutions exhibit superior resilience against physiological electromyography artifacts

Conclusion: Continuous wavelet graph transformers offer a viable pathway toward real-time wearable clini

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